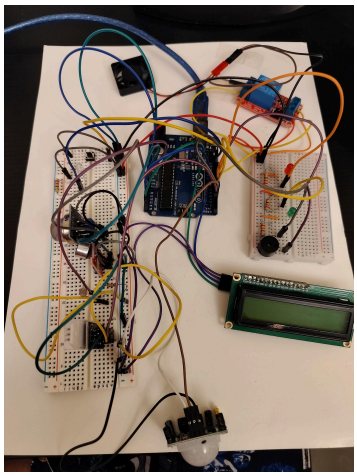
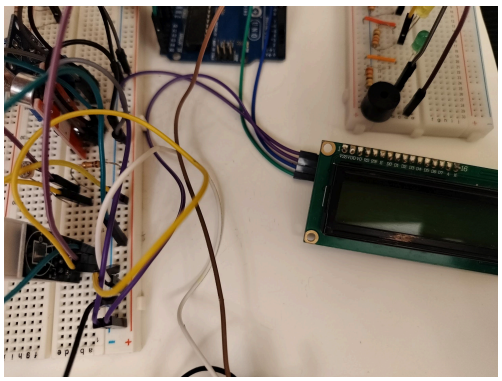
MindPulse Arduino
Pin Connections

// MindPulse is context-aware: Button choses bw/ Modes [EXAM--LECTURE--GROUP]
//
//DHT(T&H)--> if uncomfortable: FAN ON
//LDR --> WHITE LIGHT LED
//KY-038(sound)--> contributes to SI
//HC-SR501(PIR)-->MOTION/NO MOTION Display (lcd)
//MQ-135(gases: NH4,CO2)--> FAN ON & contributes to CLI
//SI/CLI (Stress / Cognitive Load Index)--> G-Y-R LED + buzzer [0-50-75-100]
//
//SI = 0.6(noise risk)+0.25(comfort risk) + 0.15(light risk)
//CLI = 0.6(air risk)+0.25(comfort risk) + 0.15(noise risk)
// rise = raw - base; risk =map(rise,low,high,0-100);
//adaptive baseline : base(new)=(1-a)*Base(old)+ a*Current
//prediction: rate= current-previous /t; timerisk = critical threshold - current index/ rate;
//Testing? button(modes);light; motion(display);sound(clap),dht(fan)

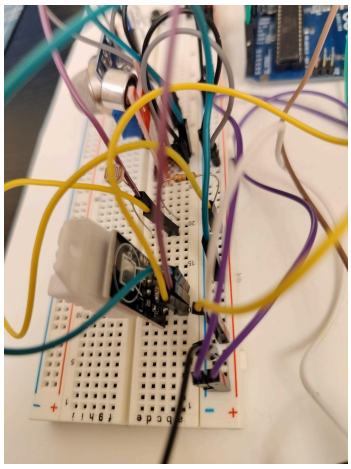


1. **LCD**



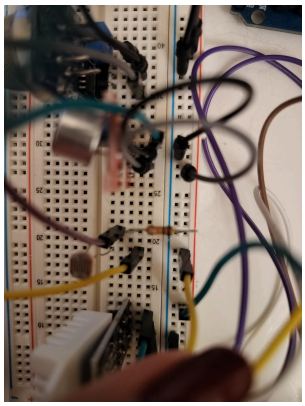
GND ——————————— GND
VCC ——————————— 5V
SDA ——————————— A4
SCL ——————————— A5

2. **DHT 22**



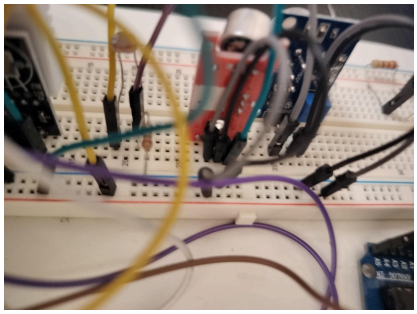
VCC/+ → 5V rail
OUT → Arduino D2
GND/- → GND rail

3. **LDR**



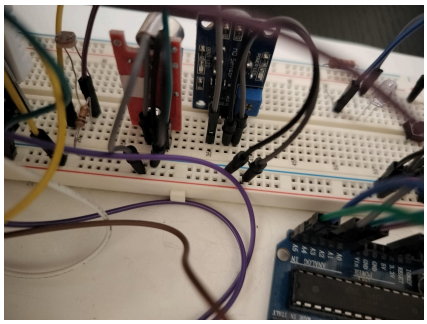
One leg - 5V
Other leg - A0 & 10 kohm to GND

4. **Sound (KY-038)**



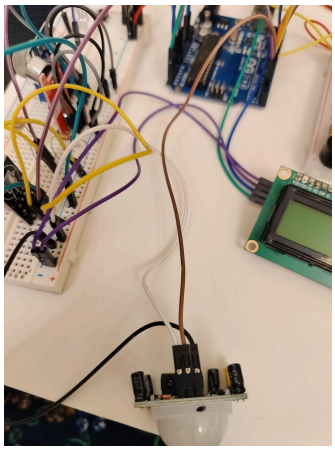
+ / VCC → 5V
G / GND → GND
AO → A1

5. MQ 135



VCC → 5V
GND → GND
AO → A2

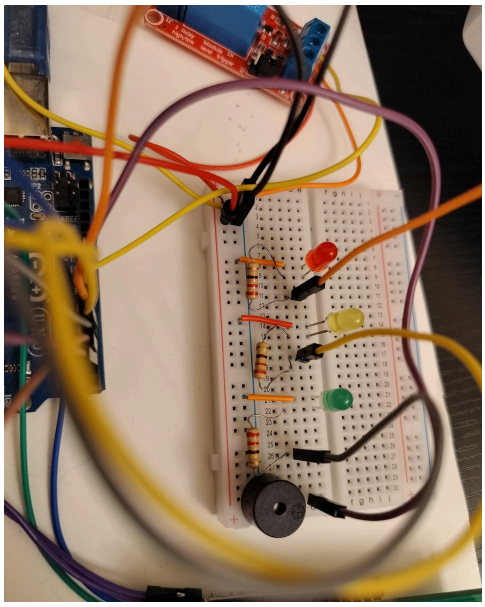
6. PIR (HC-SR501)

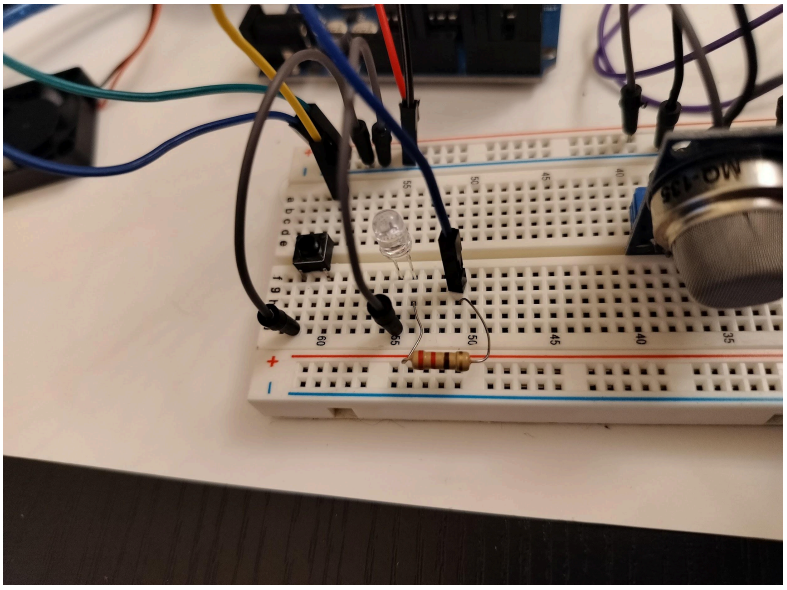


VCC → 5V
OUT → D3

GND → GND [NOTE - DOT MARKED IS GND since unlabelled]

7. LEDS (RYG + WHITE)

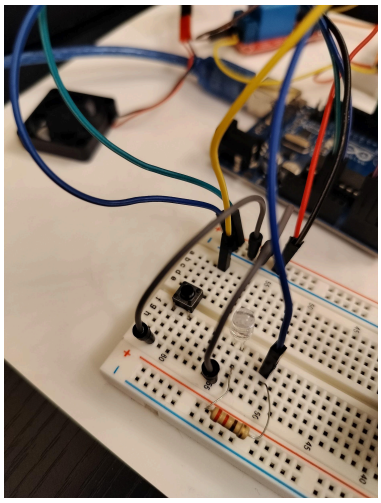




Long leg → 220 to D(X)
Short leg → GND

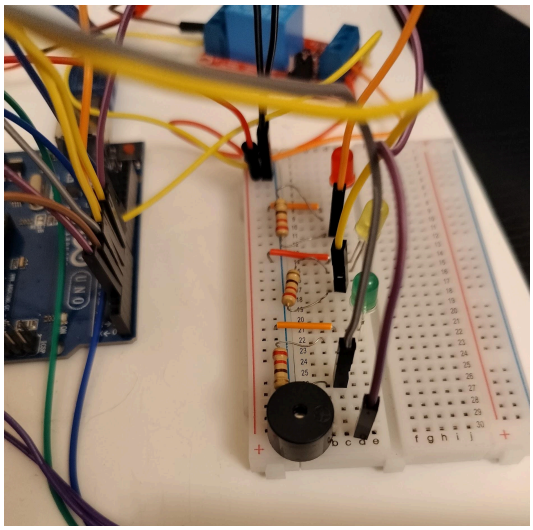
GREEN = D4; YELLOW = D5; RED = D6; WHITE = D10

8. BUTTON

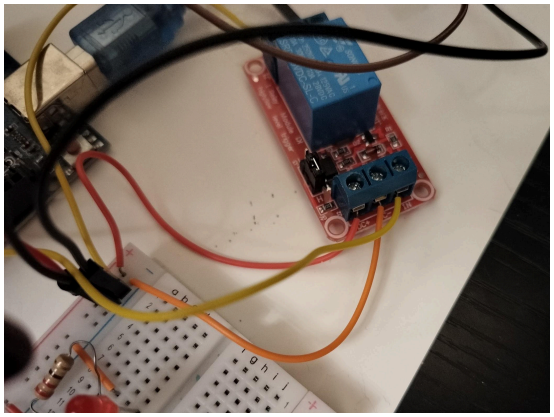


ACROSS GAP : 1 side → D7; OTHER → GND

9. BUZZER



- → GND
+ → D8



10. RELAY & FAN

RELAY TO ARDUINO :

DC+ → 5V
DC- → GND
IN → D9

—

FAN RED → Relay NO
FAN BLACK → Arduino GND
Relay COM → +5V

